MIL-T-52198C
4 April 1978
SUPERSEDING
MIL-T-52198B
25 June 1970

MILITARY SPECIFICATION

TAGLINES, CRANE AND CRANE-SHOVEL, SPRING-LOADED REEL TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers spring-loaded, independently operating, reel type taglines capable of maintaining constant tension on a clamshell bucket or other types of crane loads to prevent rotation while the load is suspended freely in the air.

1.2 Classification. Taglines shall be one of the following sizes, as specified (see 6.1):

Size 50 - For use with 1/2 cubic yard (cu yd) capacity bucket.
Size 75 - For use with 3/4 cu yd capacity bucket.
Size 150 - For use with 1 and 1-1/2 cu yd capacity buckets.
Size 200 - For use with 2 cu yd capacity bucket.
Size 350 - For use with 2 cu yd up to 3-1/2 cu yd capacity buckets.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-P-455 - Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
PPP-B-601 - Boxes, Wood, Cleated Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping Fiberboard.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standard, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 Description. Taglines shall be of the independently operating, automatic reel type with an adjustable tension mechanism permitting desired tension to be attained by manually reversing the rotation of the tagline reel, or with desired tension preset during fabrication. The taglines shall consist essentially of a reel attached to one or more helical spring-loaded cylinders and a fairlead consisting of two sheaves, a bell cable guide, or a spiral spring-loaded reel and a fairlead consisting of two sheaves and two rollers. The taglines shall be capable of being mounted on either side of the crane boom, inside the crane boom, or on the gantry frame of the crane or crane-shovel. Each tagline shall be complete and ready for operation with the reel filled with wire rope of proper size and length and with all parts necessary for installation.

* 3.2 First article. When specified (see 6.1), the contractor shall furnish one complete tagline for first article inspection and approval (see 4.2.1 and 6.3).

* 3.3 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials in the manufacturing of the taglines as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Recycled and recovered raw materials (such as glass, metals, paper, etc.) are substances which have passed through a process that puts the raw materials back to the original state. Virgin raw materials are produced directly from fresh substances, like virgin metal is made from ore. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. None of the above shall be interpreted to mean that the use of used or rebuilt products will be allowed.

* 3.4 Standard commercial product. The tagline shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product shall be included in the tagline being furnished. Standard commercial product is a product, which has been sold on the commercial market through advertisements of manufacturer's catalogs or brochures, and represents the latest production model.

3.5 Performance. Tagline, when properly installed, shall be capable of providing sufficient tension to keep the crane load from turning and fouling the crane hoisting and closing lines. Tension exerted shall be sufficient for efficient performance through all operating cycles with

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the crane or crane-shovel equipped with a boom of a length as specified in Table I, positioned at a 60° angle, and with a clamshell bucket or similar attachment operating in any position from maximum height to not less than 20 feet below ground level. Tagline shall be equipped with sufficient wire rope to permit conformance to the payout capabilities specified in Table I, with not less than two full wraps remaining on the reel.

TABLE I. Performance requirements.

Tagline size	Bucket capacity (cu yd)	Crane boom length (feet)	Approx. maximum tension required (pounds)	Minimum wire rope payout capacity (feet)
50	1/2	30	190	60
75	3/4	60	195	70
150	1 and 1-1/2	80	180	90
200	2	100	275	125
350	2 to 3-1/2	80	350	90

3.6 Reel. The tagline reel shall be of sufficient size to accommodate all necessary wire rope as specified herein. Adequate means shall be provided to anchor one end of the rope to the reel.

3.7 Fairlead. The tagline fairlead shall consist of two sheaves, a bell cable guide, or two sheaves and two guide rollers connected to the tagline mechanism by a steel bracket. Fairlead sheaves shall be heat-treated hardened steel with smooth rope grooves. Groove diameters shall be not less than the diameter of the wire rope and the inside diameter of the bell cable guide shall be not less than the diameter of the wire rope specified herein.

3.8 Wire rope. Wire rope furnished with the tagline shall be 6 by 19, filler wire construction, improved plow steel, uncoated, fiber core, right regular lay. The diameter of the wire rope shall be not less than 5/16 inch with a nominal breaking strength of not less than 8,400 pounds. The terminal (outside) end of the wire rope shall be properly seized, fitted with a thimble suitable for the wire rope specified herein, and secured by not less than two cable clamps.

3.9 Cleaning, treatment, and painting. Unless otherwise specified (see 6.1), surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.1), the unit shall be cleaned, treated, and painted in accordance with MIL-T-704, type A or C. The color of the finish coat shall be as specified (see 6.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants, such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product,

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or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.10 Identification plate. An identification plate conforming to MIL-P-514, type I, style 4 composition C, and GG-P-455, type I, grade A, class 1, shall be permanently affixed to the tagline in a conspicuous place with nonferrous screws or bolts of not less than 1/8-inch diameter. The contractor shall stamp all necessary data in the blank spaces provided for that purpose. The applicable nomenclature contained in the contract item description shall be placed in the nomenclature block. When specified (see 6.1), the plate will be furnished to the contractor by the contracting officer.

3.11 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts. The tagline shall be lubricated prior to delivery. This lubrication shall include filling tagline barrels when required by design. The tagline shall be conspicuously tagged to identify the lubricants and their temperature range. When specified (see 6.1), the tagline shall be lubricated with military lubricants and required military lubricants shall be as specified by the procuring activity.

3.12 Repair parts and maintenance tools. When specified (see 6.1), repair parts and maintenance tools shall be furnished as specified by the procuring activity.

* 3.13 Workmanship.

3.13.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.13.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

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3.13.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.13.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.13.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Examination. Each tagline shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Non-compliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

* 4.2.1 First article inspection. First article inspection shall be performed on one tagline when a first article sample is required (see 3.2). This inspection shall include the examination of 4.2 and the tests of 4.3. The first article may be a standard production item from the

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contractor's current inventory provided the tagline meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining taglines to be furnished under the contract.

4.3 First article tests. The first article shall be tested as specified herein. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all taglines until corrective action has been taken. The tagline shall be mounted on a crane or crane-shovel of proper size and which has been rigged to clamshell digging operations. The tagline, properly tensioned and in normal working condition, shall be installed on the crane boom at a height of approximately 9 feet from the superstructure connection or higher, if necessary, so as not to interfere with the crane operator's view of the work. With the crane boom positioned at a 35° angle, the clamshell bucket shall be raised to maximum height then set down or dropped on the ground in a fully opened position. The bucket shall be closed, hoisted to a height of not less than 15 feet, and simultaneously swung at least 45°. The loaded material shall be dropped, the bucket swung to its original position, and the cycle repeated for not less than 10 times. This operation shall be repeated for not less than 10 times with the crane boom raised in 10° increments until the boom reaches an angle of 65°. During these operations, the tagline function shall be closely observed to assure that no binding or chattering exists; that the tagline exerts a positive and uniform tension on the clamshell bucket under all conditions during the complete operating cycle; and that the crane hoisting and closing lines do not twist to such a degree as to cross over and foul operations.

4.4 Production unit tests. Unless otherwise specified (see 6.1), each production tagline shall be tested for proper tension and the wire rope extended full length to demonstrate free spooling and proper clearance.

4.5 Packaging inspection. The preservation-packaging, packing, and marking of the taglines shall be inspected to verify conformation to the requirements of Section 5.

5. PACKAGING

5.1 Preservation-packaging. The preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unpainted surfaces. Unpainted exterior surfaces of the taglines and component parts, including the wire rope, and unprotected metal surfaces of tools shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with MIL-P-116.

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5.1.1.2 Repair parts. Repair parts shall be preserved in accordance with the applicable submethods of preservation in MIL-P-116.

5.1.1.3 Technical publications. Technical publications for each tagline shall be preserved in accordance with MIL-P-116, method IC-3.

5.1.1.4 Consolidated packaging. Small component parts not secured to the tagline, repair parts, maintenance tools, and technical publications shall be consolidated packaged together in a close-fitting box conforming to PPP-B-636, class weather resistant.

5.1.2 Level C. Each complete tagline shall be preserved and packaged in a manner to assure protection against deterioration and damage from the supplier to the initial destination. The supplier's standard practice is acceptable when it fulfills these requirements.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Each complete tagline shall be packed in a close-fitting box conforming to PPP-B-601, overseas type; or PPP-B-621, class 2. The contents shall be cushioned, blocked, and braced to prevent movement inside the container or damaging of the contents.

5.2.2 Level B. Each complete tagline shall be packed as specified for level A, except that the boxes shall be domestic type of class 1, as applicable.

5.2.3 Level C. Each complete tagline shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of tagline required (see 1.2).
(c) When first article is not required (see 3.2).
(d) When cleaning treatment and painting shall be other than specified (see 3.9).
(e) When cleaning, treatment, and painting in accordance with MIL-T-704, type A is required (see 3.9).

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(f) Finish coat color required (see 3.9).
(g) When identification plate is required to be furnished by the contracting officer (see 3.10).
(h) When military lubricants are required and types to be furnished (see 3.11).
(i) When repair parts and maintenance tools are required and types to be furnished (see 3.12).
(j) When testing of each production tagline is not required (see 4.4).
(k) Level of preservation-packaging and level of packing required (see 5.1 and 5.2).

* 6.2 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulations (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the ASPR. The first article should be a first production item consisting of one complete tagline or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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